

Lecture - 28

Positive definite means

$$x^T A x > 0 \quad \forall x \neq 0$$

→ If A & B are positive definite.

$$+ \quad x^T A x > 0$$

$$+ \quad x^T B x > 0$$

$$\boxed{x^T (A+B) x > 0}$$

So, $A+B$ is also positive definite.

Suppose $A_{m \times n}$ is rectangular ($m \times n$)

$A^T A \rightarrow n \times n$ square symmetric

$$\Rightarrow x^T A^T A x > 0$$
$$(Ax)^T Ax > 0$$

Let $Ax = y$

$$y^T y > 0$$

$\|y\|^2$

$y^T y = 0$ in 2 case

- $x = 0$
- Nullspace of A exists ($\text{rank } A < n$)

Similar Matrices

A_n & B_n are similar when

$$B = M^{-1} A M$$

for some matrix M .

~~eg~~ $\Lambda = S^{-1} A S$

Λ is similar to A .

$$\# \quad A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & -15 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & -18 \\ 1 & 6 \end{bmatrix} = B$$

A, B, Λ are all similar

→ They have something in common

→ They have the same eigenvalues.

Proof

$$Ax = \lambda x$$

$$A = S \Lambda S^{-1}$$

$$B = M^{-1} A M$$

$$B = M^{-1} S \Lambda S^{-1} M$$

$$B = \underbrace{(M^{-1}S)} \Lambda \underbrace{(M^{-1}S)^{-1}}$$

↓
Basically eigenvectors
now become $(M^{-1}S)$
and λ 's stay the same!

'Bad' case

If $\lambda_1 = \lambda_2$ A might not be diagonalizable.

$$\underbrace{\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}}$$

lower

$$\underbrace{\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}}$$

big family

are not in the same family!!

The second matrix only has one eigenvector and is not diagonalizable.

The $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$ is a family of its own and the closest matrix to it that is diagonal is

$\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$ which is not in the same family

↓

It is also called Jordan form.

Best looking matrix
in each family.

eg more members of $\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 17 & 4 \end{bmatrix}$$

Jordan blocks

$$J_i = \begin{bmatrix} \lambda_i & & 1 \\ & \ddots & \\ 0 & & \lambda_i \end{bmatrix}$$

eg

$$\left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array} \right]$$

not similar even tho
 λ 's are same & $\#x$ is same

Jordan's theorem

Every matrix (square) is similar to a Jordan matrix J

$$J = \begin{bmatrix} \boxed{J_1} & & \\ & \boxed{J_2} & \\ & & \ddots \\ & & & \boxed{J_d} \end{bmatrix}$$

$$\boxed{\# \text{ blocks} = \# \text{ eigenvectors}}$$

$\Rightarrow$ For n distinct eigenvectors & eigenvalues, the J blocks are just the λ 's.

$$J = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = \Lambda$$

